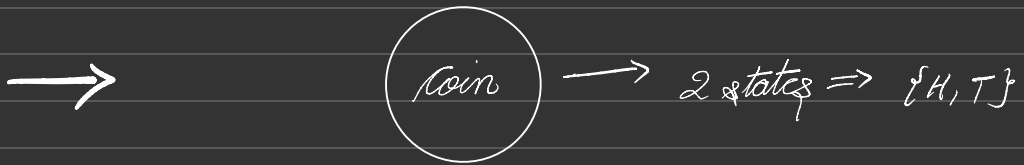
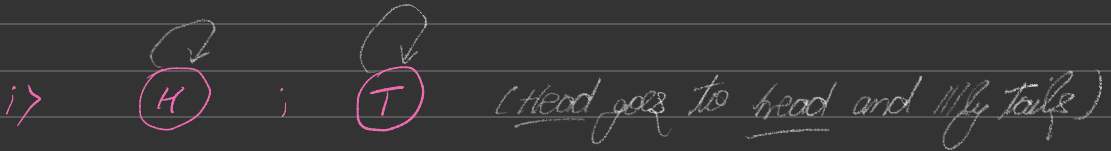


Classical Mechanics

Lecture - 1 :



Refining our own lang of motion :-

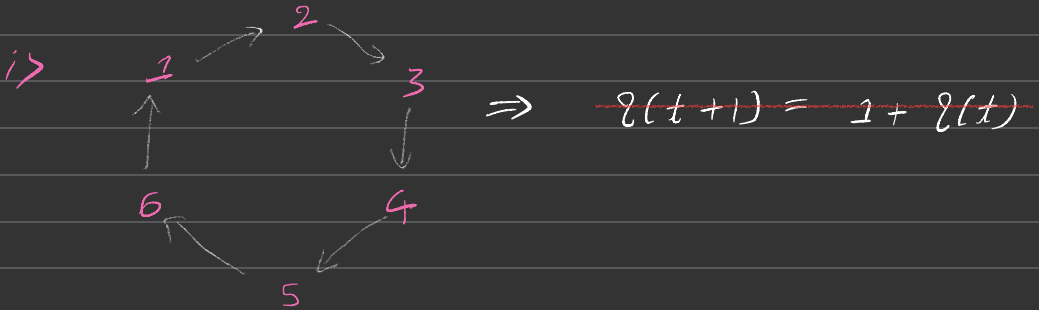
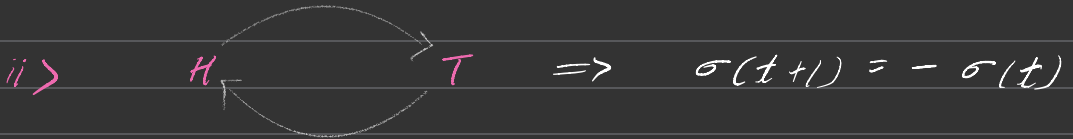


i.e. : If you start w/ H, you'll always get H and same for tails.

$$(H=1; T=-1)$$

Now, eqⁿ of motion for (i) :

$$\sigma(t+1) = \sigma(t) \quad (t \in \mathbb{Z})$$

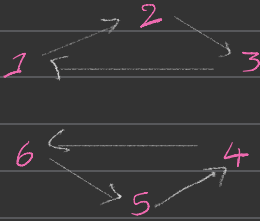


$$\Rightarrow \begin{cases} q(t+1) = 1 + q(t) & ; q(t) \leq 5 \\ q(t+1) = 1 & ; q(t) = 6 \end{cases}$$

correct?

(2 cycles)

ii >

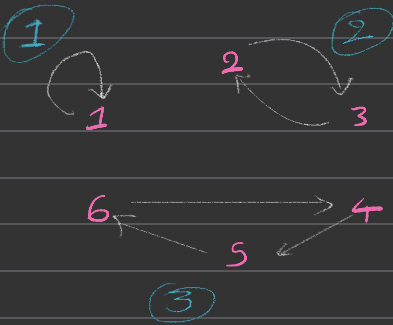


conserved quantities?

How? What does conserved q. even mean?

got it!

iii >



1, 2, 3 are conserved quantities

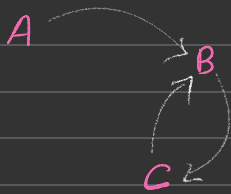
If I start w/ conserved quantity 2, I will stay on 2, even tho the state may change, 2 remains conserved.

Like 2 being mechanical energy and kinetic, potential energy being 2 states of that, interchangeable, but mech. energy remains conserved.

→ Law not allowed in classical mech.

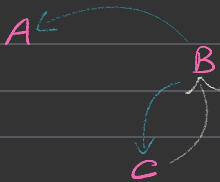
Law that you can't trace back to past.

Eg. :



completely deterministic law
(for future)

but if you reverse it to go back in past :



'B' becomes pt. of unpredictability
(undeterministic)

i.e. it's not reversible !!

∴ not allowed.

→ Principle of least action

(stationary)

for 1D

$$\frac{dV(x)}{dx} = 0$$



$$\frac{dV}{dx} \delta x = \delta V$$

($F_{\text{net}} = 0$)
(equil.)

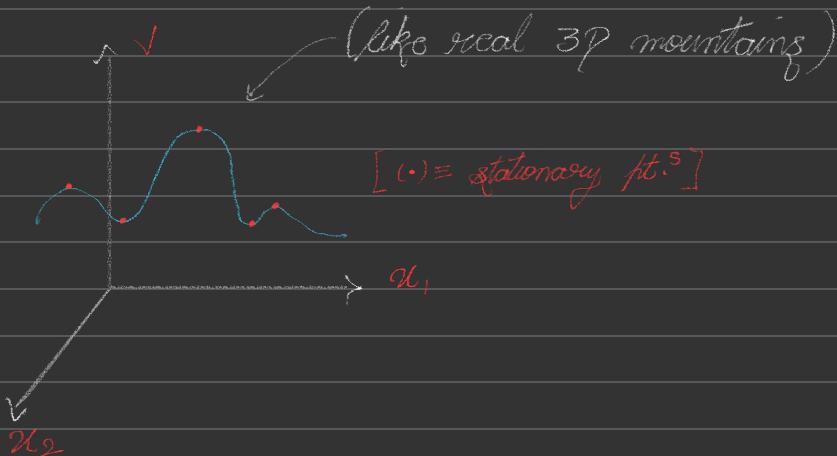
for $\geq 1D$

$$\frac{\partial V(x)}{\partial x_i} = 0$$

change in V per unit x
times small change in x

i.e. δx is the small

change in V , i.e. δV





$(\bullet) \equiv$ saddle / stationary pt.

$$\text{Here, } \delta V = \frac{\partial V}{\partial x_1} \delta x_1 + \frac{\partial V}{\partial x_2} \delta x_2$$

$\therefore$ when at equll. $\boxed{\delta V = 0}$

$\rightarrow$ Euler-Lagrange Eq.ⁿ (EL)

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{x}_i} - \frac{\partial \mathcal{L}}{\partial x_i} = 0$$

$$\mathcal{L} = T - V = \frac{1}{2} m \dot{x}_i^2 - V(x_i)$$

Using EL $\circ$ $m \ddot{x}_i + \frac{\partial V}{\partial x_i} = 0$

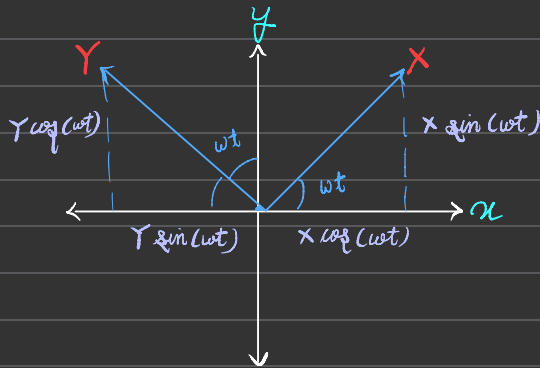
$$m \ddot{x}_i = - \frac{\partial V}{\partial x_i} = \boxed{F(x_i) = m a_i}$$

Lagrangian

This 1 fun. packages all the laws of physics for that system

It's also a really good tool for changing coordinates, as it does not depend on one, unlike eq.ⁿ of motions.

Example: X, Y rotate while x, y are fixed.



$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{x}_i} = \frac{\partial \mathcal{L}}{\partial x_i}$$

$$x = X \cos(\omega t) - Y \sin(\omega t)$$

$$y = X \sin(\omega t) + Y \cos(\omega t)$$

Here $\mathcal{L} = T$ ($\because$ no change in V)

$$\dot{x} = -X\omega \sin(\omega t) - Y\omega \cos(\omega t)$$

$$\dot{y} = X\omega \cos(\omega t) - Y\omega \sin(\omega t)$$

$\rightarrow X, Y$ vary too.

write in terms of $X = x \dots + i y \dots$
 $Y = \dots$

will solve eq. on infinite canvas and attach later.